PCCOE International Grand Challenge: 2026
Pune, India

Theme: AI for Climate Change

Title: AI-Powered Climate Action Tracker & Policy Alignment Agent for Cities

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Cities Have Climate Plans. But Who Tracks What's Actually Happening?

Fragmented Data

Climate evidence spread across disconnected sources with no common structure or tracking standard

Unverifiable Progress

Progress against specific actions is difficult to verify independently; self-reported data dominates

Late Detection

Implementation gaps identified too late for corrective interventions to be effective

Passive Dashboards

Existing tools report data without recommending action or explaining what to do next

Inaccessible Findings

Technical climate information remains opaque to citizens and non-specialist stakeholders

The gap between climate planning and verified, actionable progress is the problem we solve.

OUR SOLUTION

Evidence-to-Action System

An AI-powered system that tracks city climate commitments, identifies implementation gaps and recommends the next best interventions — grounded in verifiable, cross-source evidence.

Unified Tracking

Climate commitments across city plans, reports, satellite data, and sensors — all in one place

Cross-Validated Evidence

Evidence verified from multiple independent sources, not just self-reported data

Automated Gap Detection

Identifies divergence between planned targets and observed on-ground progress in real time

Citizen-Friendly Insights

Translates complex technical findings into accessible, understandable language for all stakeholders

WHY OUR IDEA?

Our Unique Value Proposition

We're not just tracking data; we're creating an intelligent system that transforms fragmented climate information into verifiable insights and actionable steps for cities.

Bridging the Gap

Connects ambitious climate plans with real-world implementation, overcoming data silos and lack of accountability.

AI-Driven Verification

Leverages advanced AI to cross-reference data from diverse sources, ensuring unprecedented accuracy and objectivity in progress reports.

Actionable Intelligence

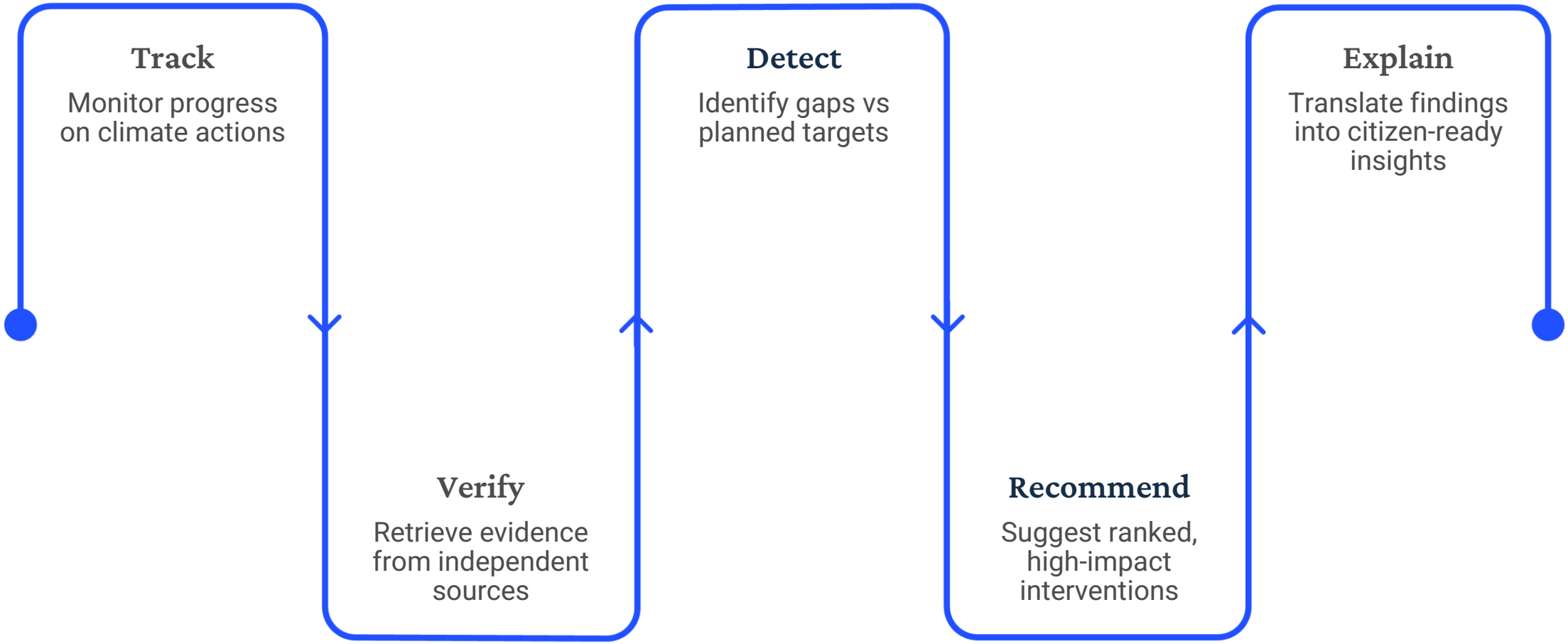
Moves beyond passive reporting to provide proactive recommendations and early warnings, enabling timely corrective interventions.

Empowering Stakeholders

Translates complex climate data into accessible formats, fostering transparency and citizen engagement in climate governance.

Five Key Objectives

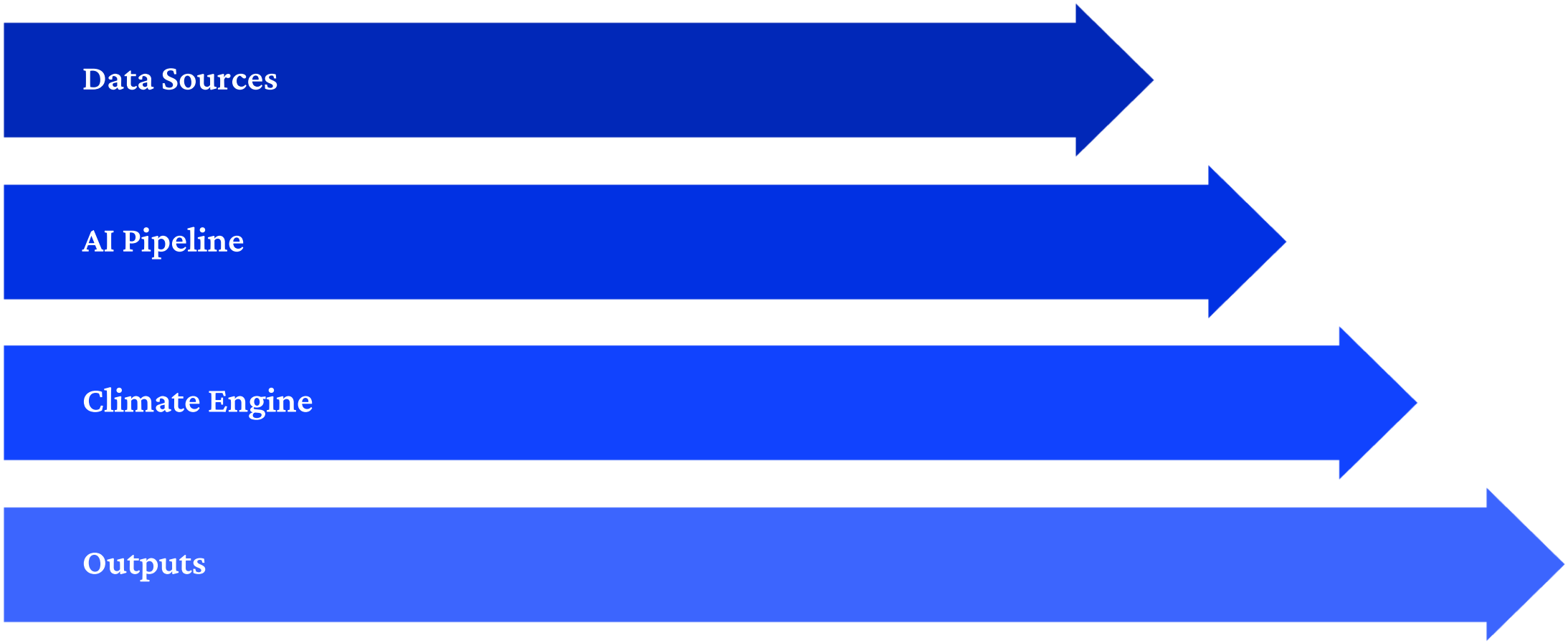
A sequential, evidence-driven workflow — each step builds on the last to move from raw data to actionable insight.



Every city commitment enters the pipeline at **Track** and emerges as an **Explained**, citizen-ready insight — with every reasoning step preserved and auditable.

System Architecture

Left-to-right data and reasoning flow — from raw, heterogeneous inputs through AI processing to structured, actionable outputs.



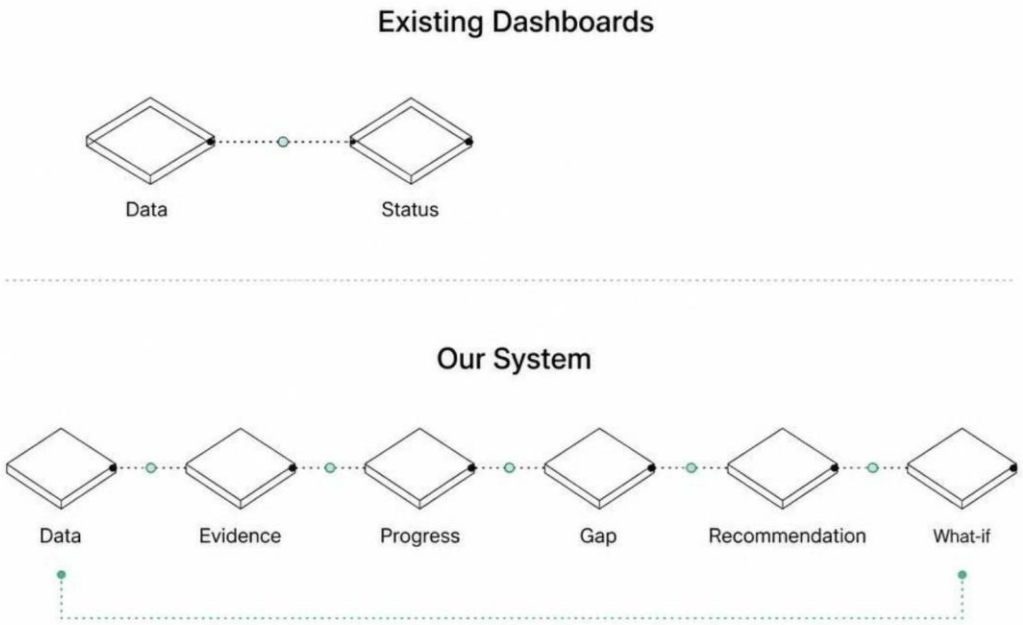
Technology Stack

- **LLM** — reasoning and explanation generation
- **RAG** (Retrieval-Augmented Generation) — evidence-grounded responses
- **Modular pipeline** — each stage independently replaceable

Design Principle

Each module is independently replaceable. Swap a satellite provider, upgrade the LLM, or extend the climate engine — without rebuilding the entire system.

Not Just a Dashboard — An Evidence-to-Action Agent



Five Key Differentiators

1. **Real-World Impact**
   Helps cities take measurable action against climate change
2. **Unique AI Approach**
   Combines LLM, retrieval, satellite data, and real-time sensors
3. **Action-Oriented**
   Identifies gaps and recommends high-ROI solutions
4. **Transparency & Trust**
   Shows sources behind every score — credible for governments and citizens alike

Every step in the reasoning chain is preserved and traceable. The system does not just report — it reasons, flags, and proposes next steps grounded in cross-validated evidence.

EXPECTED OUTCOMES & IMPACT

Built to Drive Measurable Climate Action

□ Tracks City Climate Actions

Using city plans, reports, satellite data, and real-time sensors for continuous, multi-source monitoring

Scores Progress

Across transport, energy access, and informal settlements — with sector-specific granularity

Identifies Gaps

Recommends high-ROI climate interventions and runs what-if simulations for decision-makers

Citizen Dashboards

Transparent, trustworthy, source-attributed insights accessible to all stakeholders — not just experts

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What Makes It Different

- Only system that combines **LLM + RAG + satellite + sensor** data in one climate action pipeline
- Goes beyond tracking — **identifies gaps and recommends solutions** with simulated impact
- Source attribution at every step — **credible for governments and citizens**

> From fragmented climate data to verified, actionable insight — this is the infrastructure cities need to turn plans into progress.

Driving Climate Action: Our Call to Cities

The time for verifiable, AI-driven climate action is now.

Key Takeaways

Bridge the Gap

Our system transforms fragmented data into verifiable progress, closing the gap between climate plans and on-ground implementation.

Actionable Intelligence

Leverage AI to detect gaps, recommend high-impact interventions, and simulate outcomes for smarter decision-making.

Transparency & Trust

Build public confidence with source-attributed insights and citizen-friendly dashboards, fostering accountability.

Next Steps

- Schedule a detailed demo for your city
- Discuss a potential pilot program and implementation roadmap

We are ready to partner with visionary cities to turn climate commitments into measurable, real-world impact.

Summary

The Problem

Climate plans are often fragmented, hard to verify, and difficult to translate into measurable action.

Our Approach

We combine AI, RAG, satellite data, and sensors to turn scattered signals into trusted climate intelligence.

Key Features

Continuous monitoring, progress scoring, gap detection, intervention recommendations, and citizen dashboards.

Expected Impact

Help cities move from commitments to accountable execution with faster, smarter, and more transparent climate action.